Saiyed Shizain | CU24250233 | Roll: 48
BTech CSE (A) | 3rd Yr - 5th Sem | 2024-2028

Full-Stack Web Development — BTCS303P

LAB SHEET — 4: React Routing & Global State (ShopReact)

Setup — routes, data, state

Aim. Map URLs to pages with react-router-dom and share cart state through context.

Theory. BrowserRouter matches Routes; :id params drive detail views; context avoids prop-drilling.

```
$ npm install react-router-dom
$ npm run build
✓ 35 modules transformed.
✓ built in 737ms
```

```
$ curl -s http://127.0.0.1:8814/ | grep -o '<title>[^<]*</title>'
<title>ShopReact — Saiyed Shizain</title>
(index serves HTTP 200)
```

Source — src/data/products.js

Aim. Static catalogue of three products used by every view.

```
// Lab Sheet 4 — Saiyed Shizain: static catalogue data.
export const products = [
  {
    id: 1,
    name: "Wireless Headphones",
    price: 2499,
    category: "Electronics",
    description: "Comfortable wireless headphones with clear sound."
  },
  {
    id: 2,
    name: "Running Shoes",
    price: 1999,
    category: "Shoes",
    description: "Lightweight shoes designed for daily running."
  },
  {
    id: 3,
    name: "Smart Watch",
    price: 3499,
    category: "Accessories",
    description: "Smart watch with fitness and notification features."
  }
];
```

Source — src/context/CartContext.jsx

Aim. CartProvider with add/remove; useCart hook (ASI return bug from reference fixed).

```
import { createContext, useContext, useState } from "react";

const CartContext = createContext();

export function CartProvider({ children }) {
  const [cart, setCart] = useState([]);

  const addToCart = (product) => {
    setCart((prev) => [...prev, product]);
  };
```

```
  const removeFromCart = (id) => {
    setCart((prev) => prev.filter((item) => item.id !== id));
  };

  return (
    <CartContext.Provider value={{ cart, addToCart, removeFromCart }}>
      {children}
    </CartContext.Provider>
  );
}

// Fixed vs reference: `return` on its own line would return undefined
// (automatic semicolon insertion), so the hook is kept on one line.
export function useCart() {
  return useContext(CartContext);
}
```

Source — src/components/Navbar.jsx

Aim. Shared nav with live cart count from global state.

```
import { Link } from "react-router-dom";
import { useCart } from "../context/CartContext";

// Top navigation shared by every route.
function Navbar() {
  const { cart } = useCart();
  return (
    <nav className="navbar">
      <h2>ShopReact — Saiyed Shizain</h2>
      <div>
        <Link to="/">Home</Link>
        <Link to="/products">Products</Link>
        <span className="cart-count">Cart: {cart.length}</span>
      </div>
    </nav>
  );
}

export default Navbar;
```

Source — src/components/ProductCard.jsx

Aim. Card dispatching addToCart for its product.

```
import { useCart } from "../context/CartContext";

// Catalogue card that adds its product to the global cart.
function ProductCard({ product }) {
  const { addToCart } = useCart();
  return (
    <article className="product-card">
      <h3>{product.name}</h3>
      <p>₹{product.price}</p>
      <button onClick={() => addToCart(product)}>Add to Cart</button>
    </article>
  );
}

export default ProductCard;
```

Source — src/components/Loading.jsx

Aim. Loading fallback UI.

```
// Fallback UI while product data is being fetched.
function Loading() {
  return (
    <div className="loading">
      <p>Loading products...</p>
    </div>
  );
}

export default Loading;
```

Source — src/pages/Home.jsx

Aim. Landing page at /.

```
// Landing page of the single-page shop.
function Home() {
  return (
    <main className="page">
      <h1>Welcome to ShopReact</h1>
      <p>A simple React single-page shopping application.</p>
    </main>
  );
}

export default Home;
```

Source — src/pages/Products.jsx

Aim. Grid catalogue (reference import path ../data corrected).

```
import { Link } from "react-router-dom";
import { products } from "../data/products.js";

// Fixed vs reference: import path corrected from "../data" to
// "../data/products.js" (the reference path does not exist).
function Products() {
  return (
    <main className="page">
      <h1>Products</h1>
      <div className="product-grid">
        {products.map((product) => (
          <article className="product-card" key={product.id}>
            <h3>{product.name}</h3>
            <p>{product.category}</p>
            <strong>₹{product.price}</strong>
            <Link to={`/products/${product.id}`}>View Details</Link>
          </article>
        ))}
      </div>
    </main>
  );
}

export default Products;
```

Source — src/pages/ProductDetails.jsx

Aim. Detail view via useParams :id with not-found fallback.

```
import { Link, useParams } from "react-router-dom";
import { products } from "../data/products.js";

// Detail view driven by the :id route parameter.
function ProductDetails() {
  const { id } = useParams();
  const product = products.find((item) => item.id === Number(id));
  if (!product) {
    return <h2>Product not found</h2>;
  }
  return (
    <main className="page">
      <h1>{product.name}</h1>
      <p>{product.description}</p>
      <h2>₹{product.price}</h2>
      <p>Category: {product.category}</p>
      <Link to="/products">← Back to Products</Link>
    </main>
  );
}

export default ProductDetails;
```

Source — src/pages/NotFound.jsx

Aim. Catch-all 404 page.

```
import { Link } from "react-router-dom";

// Catch-all 404 page for unknown URLs.
function NotFound() {
  return (
    <main className="page">
      <h1>404</h1>
      <h2>Page Not Found</h2>
      <p>The page you requested does not exist.</p>
      <Link to="/">Go Home</Link>
    </main>
  );
}

export default NotFound;
```

Source — src/App.jsx

Aim. BrowserRouter route map including * → NotFound.

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Navbar from "./components/Navbar";
import Home from "./pages/Home";
import Products from "./pages/Products";
import ProductDetails from "./pages/ProductDetails";
import NotFound from "./pages/NotFound";

// Route map: / → Home, /products → catalogue,
// /products/:id → details, * → 404.
function App() {
  return (
    <BrowserRouter>
      <Navbar />
```

```
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/products" element={<Products />} />
        <Route path="/products/:id" element={<ProductDetails />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
    </BrowserRouter>
  );
}

export default App;
```

Source — src/main.jsx

Aim. Root render wrapped in CartProvider.

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import App from "./App.jsx";
import { CartProvider } from "./context/CartContext";

createRoot(document.getElementById("root")).render(
  <StrictMode>
    <CartProvider>
      <App />
    </CartProvider>
  </StrictMode>
);
```

Deviations from the reference (documented)

1. Products.jsx/ProductDetails.jsx imported from nonexistent '../data'; corrected to '../data/products.js' with the new data file. 2. useCart had 'return' on its own line (returns undefined via ASI); collapsed to one line. 3. Navbar shows live cart count as visible proof of global state.

Conclusion. Install, build (35 modules) and serve verified with genuine output. — Saiyed Shizain